

Public Payments Edge Assessment Brief

Organization: HelioPay Retail Group

Scenario: Public Payments Edge

Classification: synthetic test evidence only.

System summary: Internet-facing checkout and card-payment API that terminates public traffic at a load balancer and routes into payment authorization services.

Risk goal: Critical exposed findings near public ingress.

Key findings: TLS 1.0 listener on public checkout fallback; TLS 1.2 RSA key exchange at payment API; RSA-2048 JWT validation in authorization service; ECDH between gateway and ledger adapter.

Expected scan behavior: High exposure, CRITICAL remediation for TLS 1.0, TLS 1.2 RSA key exchange, RSA-2048 JWT validation, and ECDH service-to-service handshakes.

Board concern: card payment ingress still exposes classical public-key cryptography to harvest-now-decrypt-later collection.

Primary action: remove TLS 1.0, disable RSA key exchange suites, migrate JWT verification to ML-DSA-capable trust path, and validate TLS 1.3 hybrid readiness.

Synthetic evidence only: all domains use example.com and IP addresses use RFC 5737 documentation space.

No credentials, private keys, tokens, real customer data, or live infrastructure references are included.